DATASET REPORT

1. About

Title:

Egocentric Navigation Robot Gestures (EgoNRG): An Egocentric Dataset for Hand-Arm Segmentation and Gesture Classification for Real-World Human-Robot Teaming Applications

Date:
7/15/2025

Authors:
Frank Regal
Nuclear and Applied Robotics Group, The University of Texas at Austin
(fregal@utexas.edu)

Sanat Nair
Nuclear and Applied Robotics Group, The University of Texas at Austin
(sanatnair@utexas.edu)

Asha Karmakar
Nuclear and Applied Robotics Group, The University of Texas at Austin
(asha.karmakar@utexas.edu)

Mitch Pryor
Nuclear and Applied Robotics Group, The University of Texas at Austin
(mpryor@utexas.edu)

2. Research Areas

Discipline:
Engineering, Computer Science, Social Science

Field of study:
Human-Robot Interaction (HRI), Human-Robot Collaboration

Theme of study:
Egocentric Semantic Segmentation, Egocentric Gesture Classification, Natural Command Interpretation, Egocentric Perception, Joint Hand and Arm Segmentations

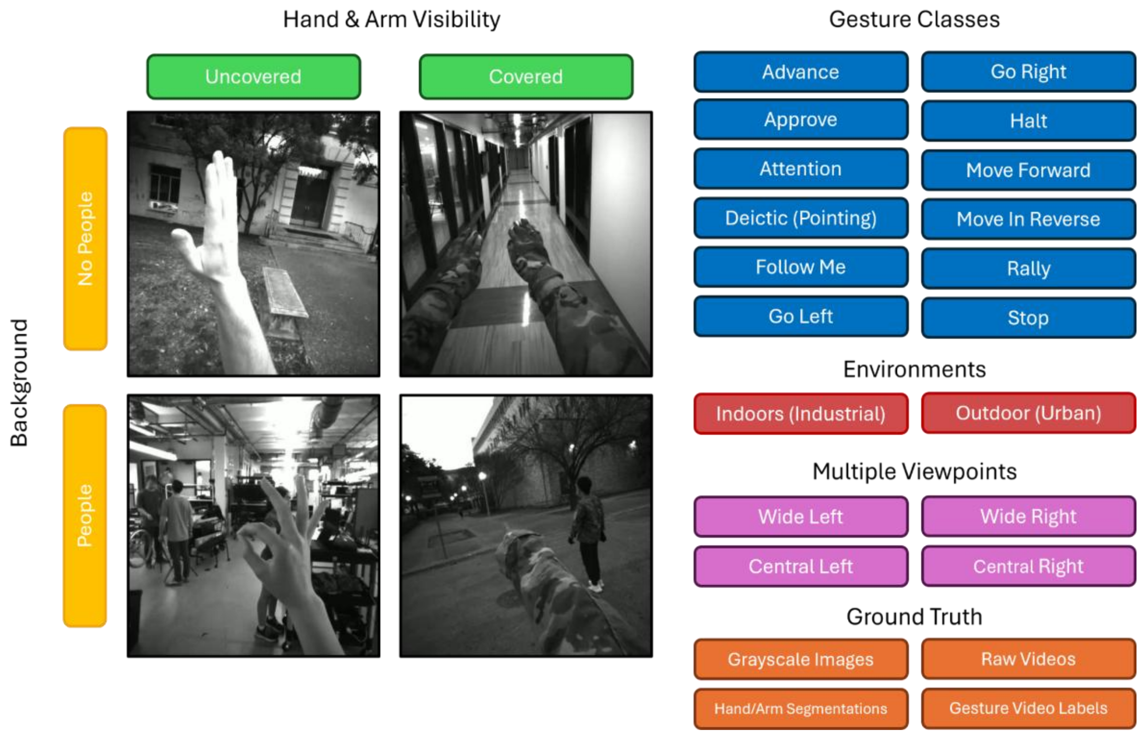
3. Dataset Description

The Egocentric Navigation-based Robot Gesture (EgoNRG) dataset is a comprehensive resource designed to advance Human-Robot Interaction (HRI) in industrial, military, and first response settings. It comprises approximately 2,500 raw video recordings and 131,000 annotated images, identifying and segmenting left and right combined hand and arm movements from head-mounted monochrome camera perspectives (positioned on each participant's head: wide left, central left, central right, and wide right) in both indoor (offices, robotics labs, commercial hallways, and foyers) and outdoor (urban settings featuring people, vegetation, buildings, vehicles, and other dynamic elements) spaces with varying lighting conditions. Thirty-two participants (14 females / 18 males) performed 12 navigation-related non-verbal gestures in total. The participants were split into 4 groups of 8, with each group performing 3 gestures.

A novelty of the EgoNRG dataset is its attention to real-world complexity, which is often seen in hazardous industrial and military settings. Half of the dataset was collected with participants' hands and arms fully covered by gloves and long-sleeved clothing, including solid, dull-colored clothing mimicking Flame-Resistant Clothes (FRC) and multiple-pattern military camouflage uniforms, closely simulating hazardous operational environments. Unlike most egocentric datasets, which typically focus on bare hands and segment only the hands, EgoNRG provides joint hand and arm segmentations for both bare and covered hands and arms, directly addressing the challenge of recognizing and segmenting limbs obscured by industrial attire—a critical capability for practical deployment in industrial and defense applications.

The dataset also introduces significant background complexity by including scenes both with and without bystanders. This diversity ensures that models trained on EgoNRG can reliably distinguish the camera wearer's hands and arms from those of others in varied, cluttered backgrounds—a challenge often overlooked in existing datasets. Traditional egocentric gesture datasets suffer from an additional critical limitation: they capture data from a single camera perspective, severely restricting their applicability to headsets with different camera mounting configurations. EgoNRG directly addresses this challenge through its innovative multi-viewpoint collection methodology. By simultaneously recording each gesture from four strategically positioned head-mounted cameras (wide left, central left, central right, and wide right), the dataset captures the complete gestural information space across a wide angular range. This comprehensive perspective coverage ensures that models trained on EgoNRG possess the viewpoint robustness necessary to perform effectively on any head-mounted system, regardless of its specific camera placement geometry—dramatically improving transferability to existing off-the-shelf head-mounted headsets with monochrome cameras and future headsets that have yet to be developed.

Of the twelve gestures included, eleven are derived from the Army Field Manual, while one represents generic pointing (deictic) gestures. These gestures were selected for their broad applicability across domains where non-verbal navigation commands are essential for controlling semi-autonomous ground vehicles and multi-agent robotic systems. By integrating multi-view egocentric video, challenging appearance variations, complex backgrounds, and multimodal data, the EgoNRG dataset provides a practical resource that can be used for segmentation research as well as gesture classification for HRI in robot navigation and fine-grained hand/arm segmentation in real-world environments. It is intended to drive progress toward robust, deployable HRI systems capable of operating in complex, high-stakes environments.



4. Dataset Contents

- **~2,500 Videos (Multi-Perspective Raw Camera Sensor Data)**
    - Unprocessed footage for custom preprocessing and experimentation was captured across 32 participants (14 Females / 18 Males), each performing 12 navigation-related non-verbal gestures, captured from four time-synchronized, head-mounted monochrome camera perspectives (positioned on each participants head wide left, central left, central right, and wide right) in both indoor (offices, robotics labs, commercial hallways, and foyers) and outdoor (urban settings featuring people, vegetation, buildings, vehicles, and other dynamic elements) spaces with varying lighting conditions. Captured and recorded using the 4 Monochrome Visible Light Cameras (VLCs) attached to the HoloLens 2 headset
- **~131K Annotated Images (Ground Truth Masks)**

    - Binary masks aligned with image frames. Annotations include “Left Limb” and “Right Limb” binary segmentation images for every frame in all videos. Each image was annotated using a semi-supervised approach.

- **~45GB Total Dataset Size**
    - 8.5 GB - Images
    - 0.5 GB - Annotations
    - 34.5 GB - Mask Images
    - 1.2 GB - Videos

- **Metadata**
    - This includes class annotation files for both raw video and audio files, dataset description file, and structured schema file.

- **IRB Consent Documentation**
    - Research Information Sheet, IRB approval ID STUDY00000278-MOD10.

- **GitHub Repository**
    - Includes dataset semi-supervised segmentation approach along with trained segmentation and classification models used for publication evaluation. Available at: https://github.com/UTNuclearRobotics/EgoNRG

5. Collection Hardware

Type:
Microsoft HoloLens 2 4x Monocular Visible Light Cameras (VLCs)

Purpose:
Capture and record egocentric point of view gestures

Adaptations:
No hardware modifications: all data collected using native HoloLens 2 sensors

6. Collection Software

- Custom Unity application for Microsoft HoloLens 2 was built for access to sensor streams
- Client codebase for external server was developed for data synchronization and writing.
- Data capture was controlled using a custom codebase that used ROS in conjunction with a modified hl2ss codebase that allowed researchers to start and stop recording of gesture sequences remotely via the HoloLens 2 interface.

7. Research Methods

Experiment Settings

- *Collection Style:*
  Physically in person
- *Geographical Location:*
  The University of Texas at Austin
- *Environment Description:*
  - Indoor (offices, laboratories, commercial building hallways, foyers, and university classrooms)
  - Outdoor (urban settings featuring people, vegetation, buildings, vehicles, and other dynamic elements)
    - morning, afternoon, and dusk
    - clouds, sunshine, shaded areas
- *Sessions:*
  Each participant completed 4 distinct sessions (one per condition combination) across two settings (indoor and outdoor).
- *Number of Sessions:*
  32 participants × 4 sessions x 2 settings = 256 total sessions
- *Duration of Sessions:*
  ~ Five minutes
- *Subjects per Session:*
  One
- *Subject Tasks:*
  Participants performed 4 predefined gestures under each session, including pointing, waving, beckoning, and directional signals. These were captured with synchronized visual sensors.
- *Study Conditions:*
  1. People are present in the background with the participants wearing arm and hand coverings (either solid color or camouflage long sleeves & gloves)
  2. People are present in the background with the participant *NOT* wearing arm and hand coverings (either solid color or camouflage long sleeves & gloves)
  3. *NO* people present in the background with the participant wearing arm and hand coverings (either solid color or camouflage long sleeves & gloves)
  4. *NO* people present in the background with the participant *NOT* wearing arm and hand coverings (either solid color or camouflage long sleeves & gloves)
- *Trials/Tasks per Session:*
  Four
- *Duration of Trials:*
  ~ Five Seconds

- *Sensor Setup and Calibration:*
  All HoloLens 2 sensors (4 VLCs Cameras) were calibrated using default factory settings; synchronized data collection was ensured using custom script.

- *Sample Images Collected:*

Data Collection Description

- *Environmental Data:*
  Microsoft HoloLens 2 recorded egocentric video and sensor time metadata.

- *Human Subjects Data:*
  Each participant was recorded in sessions involving gesture and voice command execution. Data includes video from the HoloLens 2's VLC cameras and synchronized audio.

- *Data Dictionary:*
  A structured schema includes fields for participant ID, environment type (indoor/outdoor), clothing condition (arm & hand covering/no arm & hand covering), presence of bystanders (people/no people), and ground truth segmentation files.

8. Social Sciences and Humanities Metadata

Time Method:
This data was collected in an episodic manner with each participant.

Collector Training:
Participants were given demonstrations of each gesture that was to be performed at the very beginning of the study and throughout the study if participants needed assistance.

Collection Mode:
Observational methods via the Microsoft HoloLens 2 VLC camera streams were used to gather data from participants.

Human Subjects:
The following is information about the individuals who participated in the dataset collection:

- *Age:*
  Adults aged 19-55 years old
- *Gender:*
  14 females / 18 males
- *Ethnicity:*
  Diverse participant pool; not disclosed in data
- *Total Number of Participants:*
  32
- *Occupations:*
  Students, researchers, engineers, professors
- *Distribution:*
  Primarily from Austin, TX
- *Recruitment Mode:*
  IRB approved email solicitations and word of mouth
- *IRB Number and Resolution:*
  IRB #: STUDY00000278-MOD10. Approved by The University of Texas at Austin IRB
- *Protected Data:*
  All data anonymized; no personally identifying information included. Hand and arm gestures were only recorded. Nothing links the participant to the data.

9. Dataset

Organization:
The following is the dataset directory structure:

```
EgoNRG/
└── data/
    ├── annotations/
    │   ├── Indoor/
    │   │   ├── No_People/
    │   │   │   ├── Covered/
    │   │   │   │   ├── Advance/
    │   │   │   │   │   └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │   │   │   │   │       ├── 00000.json
    │   │   │   │   │       ├── ...
    │   │   │   │   │       └── XXXXX.json
    │   │   │   │   ├── .../
    │   │   │   │   │   └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │   │   │   │   │       ├── 00000.json
    │   │   │   │   │       ├── ...
    │   │   │   │   │       └── XXXXX.json
    │   │   │   │   └── Stop/
    │   │   │   │       └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │   │   │   │           ├── 00000.json
    │   │   │   │           ├── ...
    │   │   │   │           └── XXXXX.json
    │   │   │   └── Uncovered/
    │   │   │       ├── {GESTURE}/
    │   │   │       │   ├── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │   │   │       │   │   └── ...
    │   │   │       │   └── ...
    │   │   │       └── ...
    │   │   └── People/
    │   │       └── {CONDITION}/
    │   │           └── {GESTURE}/
    │   │               └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │   │                   ├── 00000.json
    │   │                   ├── ...
    │   │                   └── XXXXX.json
    │   └── Outdoor/
    │       └── {BACKGROUND}/
    │           └── {CONDITION}/
    │               └── {GESTURE}/
    │                   └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │                       └── ...
    ├── images/
    │   └── {ENVIRONMENT}/
    │       └── {BACKGROUND}/
    │           └── {CONDITION}/
    │               └── {GESTURE}/
    │                   └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │                       ├── 00000.jpg
    │                       ├── ...
    │                       └── XXXXX.jpg
    ├── masks/
    │   └── {ENVIRONMENT}/
    │       └── {BACKGROUND}/
    │           └── {CONDITION}/
    │               └── {GESTURE}/
    │                   └── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS/
    │                       ├── 00000.png
    │                       ├── 00000.vis.png
    │                       ├── ...
    │                       ├── XXXXX.png
    │                       └── XXXXX.vis.png
    ├── videos/
    │   └── {ENVIRONMENT}/
    │       └── {BACKGROUND}/
    │           └── {CONDITION}/
    │               └── {GESTURE}/
    │                   ├── {ENVIRONMENT}_{BACKGROUND}_{CONDITION}_P{PARTICIPANT_ID}_{GESTURE}_{VIEWPOINT}_YYYYMMDD_HHMMSS.mp4
    │                   └── ...
    └── imagelists/
        ├── all.txt
        ├── test.txt
        ├── train.txt
        └── val.txt
```

Path Variables:

The following is a description of the variables shown in brackets above:

- *{ENVIRONMENT}:*
  Indoors / Outdoors
- *{BACKGROUND}:*
  People (P) / No People (NP)

- *{CONDITION}:*
  Covered (S) / Uncovered (NS)
- *{PARTICIPANT ID}:*
  3-digit number assigned to each participant
- *{GESTURE}:*
  Advance, Approve, Attention, Follow Me, Go Left, Go Right, Halt, Move Forward, Move in Reverse, Rally, Stop
- *{VIEWPOINT}:*
  Wide Left Camera (LL), Central Left Camera (LF), Central Right Camera (RF), Wide Right Camera (RR)

Directory Details:
The following is a description of the top-level directories in the dataset:

- *annotations/*
  This directory contains class frame pixel segmentation and bounding box values in a *.json* file, named to match the original image filename. The pixel values describe the segmentation area contour in the format [x_1, y_1, ..., x_n, y_n], where x-coordinates start from the bottom left side of the image and increase to the right, and y-coordinates start from the top left part of the image and increase toward the bottom. The bounding box pixel values represent the coordinates for the bottom left and top right corners of the box in the format [x_{bottom}, y_{bottom}, x_{top}, y_{top}].
- *images/*
  This directory contains the set of raw images in *.jpg* format, that were extracted as frames from each video collected for each trial.
- *masks/*
  This directory contains the set of annotated images. For each raw image, there is a matching palletized image (*.png*) and segmentation overlay (*.vis.png*) image in this directory. The palletization colors are in RGB format and are the following for each class:
  - (0,0,0) – Background Label
  - (0, 0, 255) – Left Limb Label
  - (0, 255, 0) – Right Limb Label
- *videos/*
  This directory contains the set of raw videos in *.mp4* format, that were recorded during each trial
- *imagelists/*
  This directory contains text files describing the path to each image that was used for test, train, and validation splits during training

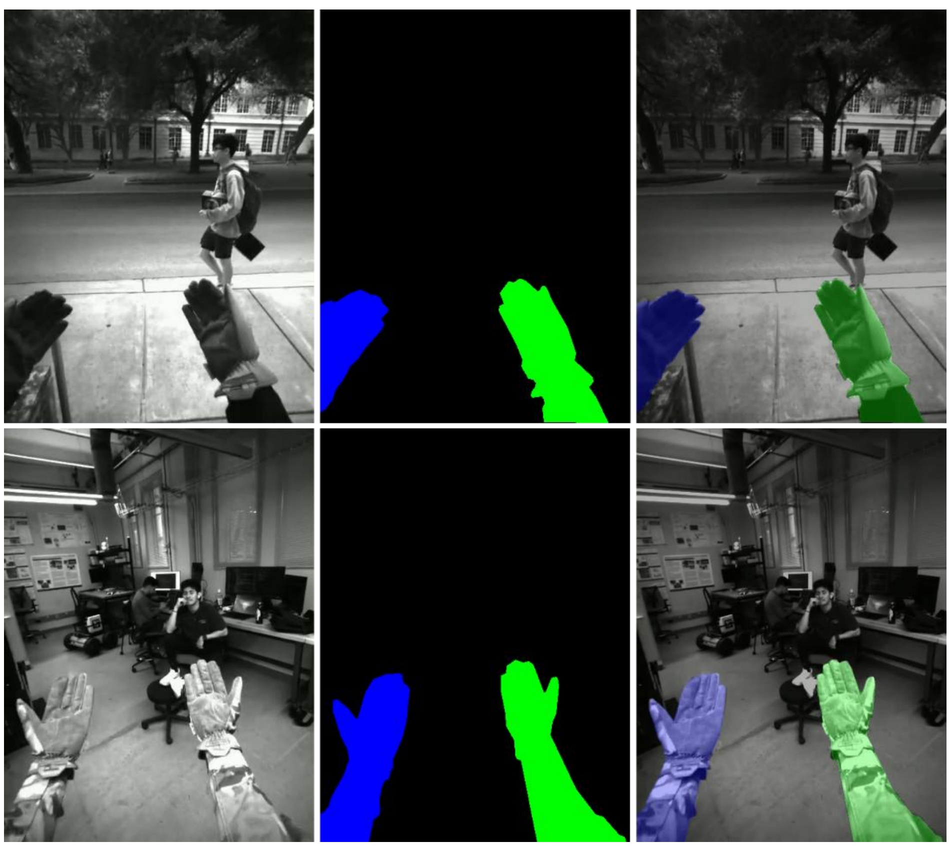
10. Annotation

Process:

The data was manually annotated by nine researchers. Three classes were assigned to each image: “left limb”, “right limb”, and background”. Human annotators were instructed to annotate each limb as the joint hand and arm for all images they could tell the hand/arm of the participant was in the image. There were three steps to the annotation pipeline. The first step for the human annotators was to review left limb and right limb bounding boxes that were automatically generated using text prompts with GroundingDINO. Once the bounding boxes for each frame were varied, these images were then automatically segmented via Segment Anything 2 (SAM2) and reassembled into videos. These videos were then manually reviewed by the annotators with a tool that played the videos back at 1 FPS and the option to manually skip through the frames of the video. For each frame in the video that had incorrect pixel segmentations, annotators flagged these frames. Annotators then manually reviewed and fixed the pixel segmentations of the frames that were flagged. Each frame’s annotation was converted to a single PNG file, where the three classes were recorded: left hand, right hand, and background.

Example Annotations:

Below is an example image highlighting two different annotations (each row). In the leftmost column is the original frame of an image. The middle column is the palletized annotation image, and the right column is the overlayed segmentation image for easy viewing.



11. Evaluation (ML Models & Software)

Details:

Two different experiments were conducted on this dataset. The first (1) experiment that was conducted was the evaluation of semantic segmentation models on the dataset images and the second (2) experiment that was conducted was the evaluation of gesture classification models on the ground truth annotated images recompiled as videos. Details about both evaluations are below:

- *Semantic Segmentation Models:*
  Four different State-of-the-Art (SOTA) semantic segmentation models were evaluated against the annotated images of the dataset. As this dataset was collected with the intention to be deployed on headsets that have limited onboard compute, three SOTA models shown to have properties of real-time inference were chosen and one baseline SOTA model for semantic segmentation was chosen. The models chosen for evaluation are listed below:
    - SegFormer
    - PPLiteSeg
    - PIDNet-S
    - BiSeNetV2

  These models were all finetuned on the dataset annotations from each of their respective checkpoints that were trained on the CityScapes dataset.

- *Gesture Classification Models:*
  Three different SOTA gesture classification models were evaluated on the annotated videos that were stitched together with the ground truth palletized images. Again, lightweight models that have the potential to perform gesture classification in real-time on edge computing devices were chosen. The following models that were used are:
    - VGG16 (fcn16)
    - VGG16 + LSTM (lstm3)
    - CSD (fc6, 16 frames)

Codebase:

The trained segmentation and classification models used for publication evaluation are available at: https://github.com/UTNuclearRobotics/EgoNRG. This repository includes:

- Preprocessing scripts
- Model training and evaluation code
- Inference pipeline
- Environment setup instructions
- README with usage examples

Results:

Results from each of the evaluations are listed below:

- *Semantic Segmentation Models:*

Model	mIoU (↑)	Precision (↑)	Recall (↑)	F1-Score(↑)	Avg. Inference Time (ms)	FPS
BiSeNetv2	-	-	-	-	-	-
PP-LiteSeg	-	-	-	-	-	-
SegFormer	-	-	-	-	-	-
PIDNet-S	-	-	-	-	-	-

- *Gesture Classification Models:*

Model	Accuracy(↑)	Precision (↑)	Recall (↑)	F1-Score(↑)	Avg. Inference Time (ms)	FPS
VGG16	-	-	-	-	-	-
VGG16+LSTM	-	-	-	-	-	-
C3D	-	-	-	-	-	-

12. Quality Control Statement

Standardized Data Collection:
All data was gathered under consistent conditions, following strict protocols for operation and sensor calibration.

Validation and Cleaning:
Automated and manual checks were conducted to detect errors, missing values, and inconsistencies.

Annotation Accuracy:
Multistep verification ensured high-quality labeling, with inter reliability checks where applicable.

Ethical Compliance:
Identifiable data was anonymized, adhering to institutional review board (IRB) guidelines.

Reproducibility:
Metadata and a structured dataset hierarchy support easy navigation and replication.

System Integrity:
Regular monitoring and recalibration-maintained sensor and robot performance.

Research Validation:
Rigorous methodologies ensured experimental reliability, with controlled conditions and standardized protocols.

Data Integrity:

Automated and manual validation processes detected inconsistencies, ensuring accuracy and completeness.

Ethical Compliance:
Data collection followed ethical guidelines, with anonymization and consent procedures in place.

System Performance:
Continual monitoring of image streams ensured proper functionality of sensors used to record the data.